



HADA: HANA Data Dependency Assistant

☒ Demo mode: Use example dependencies

Clear All

Choose demo query:

TPCDS Catalog Returns Complex Join

1 SQL Query Input

Original Query

Reset Query

```
1 SELECT cr_order_number, cr_item_sk, cs_sold_date_sk, c_customer_id, ca_city
2 FROM catalog_returns_sanitized
3 INNER MANY TO ONE JOIN catalog_sales_sanitized
4 ON cr_order_number = cs_order_number AND cs_item_sk = cr_item_sk
5 INNER MANY TO ONE JOIN date_dim ON cs_sold_date_sk = d_date_sk
6 INNER MANY TO ONE JOIN customer ON cs_bill_customer_sk = c_customer_sk
7 INNER MANY TO ONE JOIN customer_address ON c_current_addr_sk =
  ca_address_sk
8 WHERE d_year BETWEEN 2000 AND 2002
9 AND ca_country = 'United States'
10 AND d_moy >= 6;
```

Discover Dependencies

Rewritten Query

Reset Rewrite

```
SELECT cr_order_number, cr_item_sk, cs_sold_date_sk, c_customer_id, ca_city
FROM catalog_returns_sanitized
  INNER MANY TO ONE JOIN catalog_sales_sanitized
    ON cr_order_number = cs_order_number AND cs_item_sk = cr_item_sk
  INNER MANY TO ONE JOIN date_dim ON cs_sold_date_sk = d_date_sk
  INNER MANY TO ONE JOIN customer ON cs_bill_customer_sk = c_customer_sk
  INNER MANY TO ONE JOIN customer_address ON c_current_addr_sk = ca_address_sk
WHERE d_year BETWEEN 2000 AND 2002
  AND ca_country = 'United States'
  AND d_moy >= 6;
WITH HINT(DEV_DATA_DEPENDENCIES('["type": "OD", "table": "SYSTEM.customer_address", "lhs":
["ca_address_sk"], "rhs": ["ca_country"]], [{"type": "FD", "table":
"SYSTEM.catalog_returns_sanitized", "lhs": ["cr_order_number"], "rhs": ["cr_item_sk"]}]'))
```

2 Dependency Selection

Valid Dependencies

7

Functional (FD)

4

Order (OD)

3

Selected

2

Valid Dependencies

These dependencies have been validated. Click to select for query rewriting.

catalog_returns_sanitized		
☒	FD VALID MANUAL cr_order_number → cr_item_sk	Deselect
catalog_sales_sanitized		
☐	FD VALID cs_order_number → cs_item_sk	Select
customer		
☐	FD VALID c_customer_sk → c_customer_id	Select
☐	FD USER VALIDATED c_customer_sk → c_birth_country	Select Invalidate
customer_address		
☒	OD VALID MANUAL ca_address_sk ↔ ca_country	Deselect
☐	OD USER VALIDATED ca_address_sk ↔ ca_state	Select Invalidate
date_dim		
☐	OD VALID d_date_sk → d_year, d_moy	Select

Selection Summary

2 dependencies selected

OD
customer_address.ca_address_sk

FD
catalog_returns_sanitized.cr_order_number

Hint Preview

```
[{"type": "OD", "table": "SYSTEM.customer_address", "lhs":
```

Apply Rewrite

Clear Selection

Invalid Dependencies

These dependencies need validation. Mark as valid to use them.

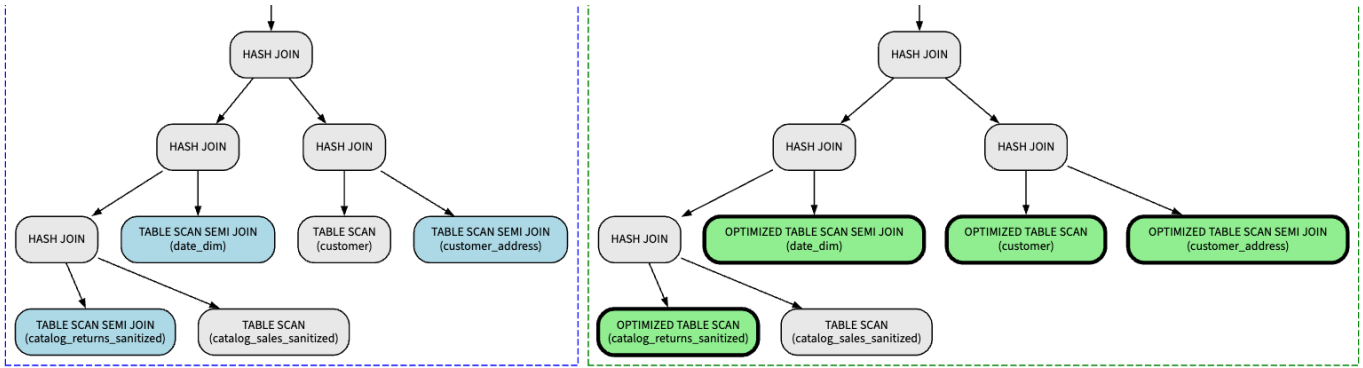
☐ FD INVALID date_dim.d_date_sk → date_dim.d_week_seq

Mark as valid

3 Query Plan Comparison

Load Query Plans



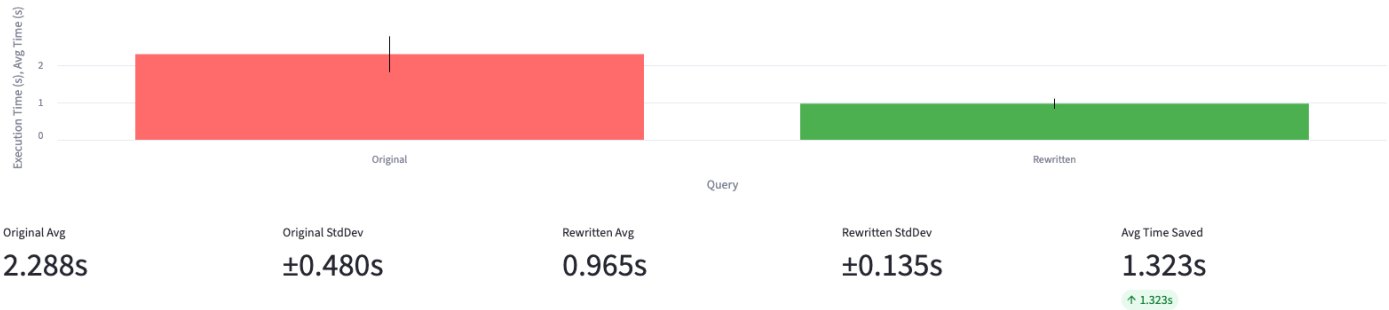


4 Performance Comparison

START

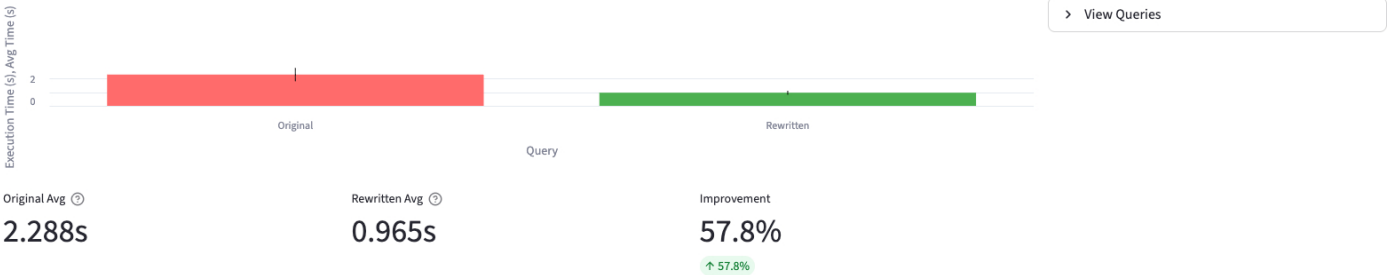
STOP

Live Benchmark Results



57.8% improvement (1.323s faster) with dependency hints! (based on 10 runs)

Benchmark History



5 Query Results Preview

Show Query Results (First 100 rows)

RESULTS IDENTICAL

Query rewrite only affects performance, not results. Both queries return the same data.

Original Query Results:

	cr_order_number	cr_item_sk	cs_sold_date_sk	c_customer_id	ca_city
0	144523	8934	2451180	AAAAAAAAABMCPFAAA	Midway
1	144523	12456	2451180	AAAAAAAAABMCPFAAA	Midway
2	167892	3421	2451245	AAAAAACDEFGAAA	Fairview
3	189034	7823	2451302	AAAAAAHIJLAAA	Springfield
4	201456	9012	2451356	AAAAAAOPQRSTAAA	Riverside
5	201456	11234	2451356	AAAAAAOPQRSTAAA	Riverside
6	223789	5678	2451412	AAAAAAVWXYZAAA	Georgetown
7	245123	2345	2451478	AAAAAAABCDEFBAA	Clinton
8	267456	6789	2451534	AAAAAAAGHJKLBAA	Franklin
9	289012	4567	2451590	AAAAAAOPQRSBAA	Madison

Rewritten Query Results:

	cr_order_number	cr_item_sk	cs_sold_date_sk	c_customer_id	ca_city
0	144523	8934	2451180	AAAAAAAAABMCPFAAA	Midway
1	144523	12456	2451180	AAAAAAAAABMCPFAAA	Midway
2	167892	3421	2451245	AAAAAACDEFGAAA	Fairview
3	189034	7823	2451302	AAAAAAHIJLAAA	Springfield
4	201456	9012	2451356	AAAAAAOPQRSTAAA	Riverside
5	201456	11234	2451356	AAAAAAOPQRSTAAA	Riverside
6	223789	5678	2451412	AAAAAAVWXYZAAA	Georgetown
7	245123	2345	2451478	AAAAAAABCDEFBAA	Clinton
8	267456	6789	2451534	AAAAAAAGHJKLBAA	Franklin
9	289012	4567	2451590	AAAAAAOPQRSBAA	Madison

6 SQL Logs

Show SQL Logs

SQL Execution Logs (showing last 20 entries, older entries are automatically pushed out)

[10:05:23]	QUERY	EXPLAIN PLAN SET STATEMENT_NAME = 'dep_check_query_123456' FOR SELECT cr_order_number, cr_item_sk...
[10:05:24]	QUERY	SELECT operator_id, parent_operator_id, operator_name, operator_details, table_name FROM SYS.EXPLAIN...
[10:05:25]	DELETE	DELETE FROM SYS.EXPLAIN_PLAN_TABLE WHERE STATEMENT_NAME = 'dep_check_query_123456'
[10:05:30]	QUERY	SELECT cr_order_number, cr_item_sk, cs_sold_date_sk, c_customer_id, ca_city FROM catalog_returns_san...
[10:05:32]	QUERY	SELECT cr_order_number, cr_item_sk, cs_sold_date_sk, c_customer_id, ca_city FROM catalog_returns_san...
[10:05:35]	BENCHMARK	Benchmark iteration 1: Original=2.34s, Rewritten=0.89s
[10:05:38]	BENCHMARK	Benchmark iteration 2: Original=2.21s, Rewritten=0.92s